

Robust Harper Stability at the Exponential Scale

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Abstract

Harper’s theorem identifies the smallest possible Hamming neighborhood of a set of prescribed cardinality. We study a stability question in a regime very different from the usual $1 + o(1)$ theory: the neighborhood may be larger than the Harper minimum by any subexponential factor. Under a linear lower bound on the neighborhood radius and a natural condition keeping the grown set below the equator, we prove that all but a prescribed fraction of the original set is covered by subexponentially many Hamming balls, each having the smallest possible radius up to $o(n)$.

The proof is information-theoretic. Near-minimal growth at one macroscopic radius first forces near-minimal growth at every smaller radius. This makes every macroscopic coordinate window entropy-typical. When a uniform point of the set is revealed one coordinate at a time, the magnitude of almost every conditional bias is consequently the same. The remaining information is the direction of that bias; its trajectory defines a canonical center and has entropy $o(n)$. A subexponential typical set of these centers, rather than a single heavy center, gives the desired cover directly.

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*A Lean 4/Mathlib formalization of the main theorem and of Harper’s theorem is available.

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1 Introduction

1.1 Harper's theorem and two notions of stability

Write $Q_n = \{0, 1\}^n$, let d be Hamming distance, and put

$$B(a, t) = \{x \in Q_n : d(x, a) \leq t\}, \quad \Gamma_r(S) = \bigcup_{x \in S} B(x, r).$$

For fixed n, k, r , let

$$V_n(k, r) = \min\{|\Gamma_r(A)| : A \subseteq Q_n, |A| = k\}.$$

We suppress the subscript n when the ambient dimension is fixed. Harper's vertex-isoperimetric theorem [1] says that this minimum is attained by an initial segment of the simplicial order. In particular, when k is the cardinality of a Hamming ball, a Hamming ball has the smallest possible r -neighborhood. We write $r_{\min}(k, n)$ for the least radius of a ball in Q_n containing at least k points.

The usual stability question asks whether a set whose boundary is only slightly larger than the minimum must itself be close to a minimizer. For the one-step vertex boundary, the theorem of Keevash and Long [2] gives such a conclusion when the multiplicative error tends to one (in its basic ball-cardinality form the error is $1 + O(1/n)$). The conclusion is closeness, in symmetric difference, to one Hamming ball. Related quantitative results were obtained by Przykucki and Roberts [3].

Here the permitted error is much larger. We allow

$$|\Gamma_r(S)| \leq 2^{o(n)} V_n(|S|, r).$$

Even a polynomial factor is already far outside a $1 + o(1)$ theorem. In this regime closeness to one ball is false: one may take many well-separated, nearly optimal pieces. The right conclusion is therefore not one-ball stability but a *structure theorem at exponential scale*: almost all of the set is a union of subexponentially many almost minimal balls.

Informal main theorem. *Fix the problem away from the three degenerate regimes: the original set has the exponential size of a ball whose relative radius stays in a compact subinterval of $(0, 1/2)$; the neighborhood radius is a positive fraction of the dimension; and the given set stays exponentially below the whole cube. If the neighborhood is within a subexponential factor of Harper's minimum, then, after discarding any prescribed fixed fraction of the points, the set is covered by subexponentially many Hamming balls. Every covering ball has the smallest radius compatible with the cardinality of the original set, up to an additive $o(n)$.*

Thus the theorem identifies the only robust obstruction to one-ball stability: subexponential fragmentation into ball-like pieces.

1.2 Formal statement

Let

$$H(x) = -x \log_2 x - (1-x) \log_2 (1-x)$$

denote binary entropy. Throughout, a term of the form $0 \log_2 0$ is interpreted as zero.

Theorem 1.1 (robust Harper stability). *Fix $\rho > 0$, $\delta_c \in (0, 1)$, $C_{\text{size}} \geq 1$, a compact interval $[\alpha_-, \alpha_+] \subset (0, 1/2)$, a function $\sigma : \mathbb{N} \rightarrow [0, \infty)$ satisfying*

$$\sigma(n) = o(n), \quad \sigma(n) \geq \log_2 n,$$

and a covering error $\varepsilon_{\text{cov}} \in (0, 1)$. Then there exists a function $\tau(n) = o(n)$, depending only on these fixed data, such that the following holds for all n .

Let $S \subseteq Q_n$, put $k = |S|$, and suppose that for some $\alpha \in [\alpha_-, \alpha_+]$ and integer $r \geq \rho n$,

$$|\log_2 k - H(\alpha)n| \leq C_{\text{size}}\sigma(n), \tag{1}$$

$$|\Gamma_r(S)| \leq 2^{\sigma(n)} V_n(k, r), \tag{2}$$

$$|\Gamma_r(S)| \leq 2^{(1-\delta_c)n}. \tag{3}$$

Then there is a set of centers $\mathcal{C} \subseteq Q_n$ with $|\mathcal{C}| \leq 2^{\tau(n)}$ such that

$$|S \setminus \bigcup_{c \in \mathcal{C}} B(c, r_{\min}(k, n) + \tau(n))| \leq \varepsilon_{\text{cov}} |S|.$$

The parameter α is the geometric scale hidden in the cardinality of S . Indeed,

$$|B(0, \lfloor \alpha n \rfloor)| = 2^{H(\alpha)n + o(n)},$$

so (1) says that S has, at exponential scale, the size of a Hamming ball of radius αn . The parameter r is the distance by which the set is grown, $\sigma(n) = o(n)$ is the permitted error in the exponent, and $\tau(n) = o(n)$ is the resulting loss in the number and radii of the covering balls.

The statement is asymptotic, but the single function τ is uniform over all sets in the class. Finitely many small dimensions may always be absorbed by enlarging τ there.

The assumptions have separate roles. The compact density interval gives uniform control of the entropy function. The lower bound $r \geq \rho n$ rules out subcubes, which can have small one-step boundary without being ball-like. The cap (3) keeps the comparison ball below the equator; after the equator, neighborhoods are close to the whole cube and near-minimality has little structural content. Finally, the conclusion must allow $2^{o(n)}$ balls: unions and chains of well-separated near-optimal pieces show that no bounded number can suffice.

1.3 Canonical centers and the reduction of the theorem

We first explain the final mechanism of the proof. This makes clear what the entropy statement below is about and why it implies a geometric cover.

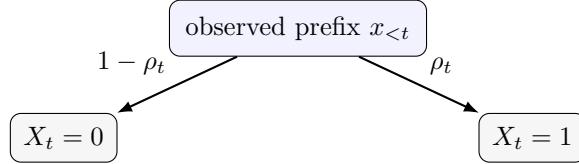
Let X be uniformly distributed on S . Revealing its coordinates in the order $1, \dots, n$ moves down the prefix tree of S . At a prefix $x_{<t}$ that occurs in this tree, define

$$\rho_t(x_{<t}) = \mathbb{P}[X_t = 1 \mid X_{<t} = x_{<t}].$$

Every prefix used here has positive probability, so the conditional probability is well defined. We extract two quantities from this branching:

$$b_t(x_{<t}) = \min(\rho_t(x_{<t}), 1 - \rho_t(x_{<t})), \quad c_S(x)_t = \mathbf{1}[\rho_t(x_{<t}) > 1/2].$$

The number $b_t(x_{<t})$ is the conditional probability that the next bit takes the less likely branch. The bit $c_S(x)_t$ records the more likely branch and becomes the t th coordinate of the *canonical center* of x . Ties are resolved in favor of 0.



$c_S(x)_t$ chooses the branch with larger probability; b_t is the smaller probability.

Figure 1: One step of the reveal process. Both quantities depend on the realized prefix $x_{<t}$.

Thus $c_S : S \rightarrow Q_n$ depends on the successive prefixes of x and on the uniform measure on S . When x is replaced by the random point X , we write

$$\rho_t = \rho_t(X_{<t}), \quad b_t = b_t(X_{<t}), \quad C = c_S(X).$$

These are random variables: different sampled points may reach different prefixes and hence have different branch probabilities and canonical centers. The variable C takes values in Q_n , and its law is the distribution of the canonical center induced by a uniform point of S . Its Shannon entropy, measured in bits, is

$$H(C) = - \sum_{z \in Q_n} \mathbb{P}[C = z] \log_2 \mathbb{P}[C = z].$$

The technical part of the paper proves the following statement. We state it now and postpone its proof until Section 11.

Proposition 1.2 (canonical-center estimate). *Fix $\rho > 0$, $\delta_c \in (0, 1)$, $C_{\text{size}} \geq 1$, a compact interval $[\alpha_-, \alpha_+] \subset (0, 1/2)$, and a function $\sigma : \mathbb{N} \rightarrow [0, \infty)$ with $\sigma(n) = o(n)$ and $\sigma(n) \geq \log_2 n$. Then there are functions*

$$g, u : \mathbb{N} \rightarrow [0, \infty), \quad e : \mathbb{N} \rightarrow [0, 1]$$

such that

$$g(n) = o(n), \quad u(n) = o(n), \quad e(n) \longrightarrow 0,$$

and the following holds. For every n , every $S \subseteq Q_n$, every $\alpha \in [\alpha_-, \alpha_+]$, and every integer $r \geq \rho n$, put $k = |S|$. If these parameters satisfy (1)–(3), the random variables X and $C = c_S(X)$ defined above satisfy

$$H(C) \leq g(n), \quad \mathbb{P}[d(X, C) > r_{\min}(|S|, n) + u(n)] \leq e(n).$$

The two conclusions have different roles. Small entropy says that almost all canonical centers belong to a subexponential list. The distance estimate says that almost every point lies in an almost minimal ball around its own center.

Lemma 1.3 (typical values of a low-entropy variable). *Let Y be a discrete random variable with Shannon entropy E , measured in bits. For every $\delta \in (0, 1)$ there is a set T of values such that*

$$|T| \leq 2^{2E/\delta}, \quad \mathbb{P}[Y \notin T] \leq \delta/2.$$

Proof. When $E = 0$, the variable is constant. Otherwise write $p(y) = \mathbb{P}[Y = y]$ and put

$$T = \{y : p(y) \geq 2^{-2E/\delta}\}.$$

The probabilities sum to one, so $|T| \leq 2^{2E/\delta}$. Moreover,

$$E = H(Y) = \sum_y p(y) \log_2 \frac{1}{p(y)}.$$

Applying Markov's inequality to the nonnegative random variable $\log_2(1/p(Y))$ gives

$$\mathbb{P}[Y \notin T] = \mathbb{P}\left[\log_2 \frac{1}{p(Y)} > \frac{2E}{\delta}\right] \leq \frac{\delta}{2}.$$

□

Deduction of Theorem 1.1 from Proposition 1.2. Apply Lemma 1.3 to $Y = C$ with $\delta = \varepsilon_{\text{cov}}$. There is a set $T \subseteq Q_n$ such that

$$|T| \leq 2^{2g(n)/\varepsilon_{\text{cov}}} = 2^{o(n)}, \quad \mathbb{P}[C \notin T] \leq \varepsilon_{\text{cov}}/2.$$

For all sufficiently large n , Proposition 1.2 also gives

$$\mathbb{P}[d(X, C) > r_{\min}(|S|, n) + u(n)] \leq \varepsilon_{\text{cov}}/2.$$

If neither exceptional event occurs, then $C \in T$ and X belongs to the ball of radius $r_{\min}(|S|, n) + u(n)$ centered at C . Hence the set of centers $\mathcal{C} := T$ covers all but an ε_{cov} -fraction of S by balls of this radius. Taking $\tau(n)$ to dominate both $2g(n)/\varepsilon_{\text{cov}}$ and $u(n)$ proves the theorem for large n . Enlarging τ in the finitely many remaining dimensions proves the stated form for every n without affecting $\tau(n) = o(n)$. □

This direct reduction is why no heavy-center lemma or greedy peeling is needed. The conclusion already permits $2^{o(n)}$ balls, so we keep every typical canonical center rather than pigeonholing the center distribution to one heavy value.

1.4 Outline of the canonical-center estimate

We now give the logical spine of Proposition 1.2. Throughout, X is uniform on the set under study. A coordinate window is a set of coordinates through which we partially observe X . Each numbered paragraph identifies its conclusion near the beginning and then explains why it is true; all asymptotic statements are made precise, uniformly over the class, in the cited later results. Steps 1–4 establish the entropy conclusion of Proposition 1.2. Step 3 also feeds Step 5, which establishes the distance conclusion; the two outputs are then combined by Lemma 1.3 as in Subsection 1.3.

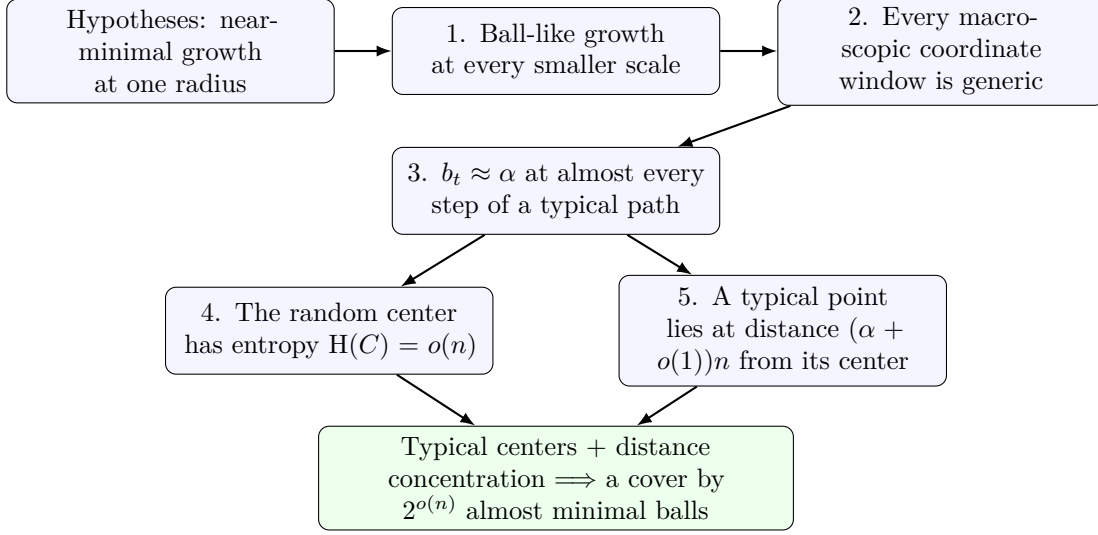


Figure 2: The proof at a glance. The five numbered boxes correspond to the five paragraphs below.

1. Near-minimal growth propagates downwards. *Conclusion of the step.* Put $\beta = r/n$. At every smaller macroscopic radius tn , the neighborhood of S is no larger, at exponential scale, than the corresponding neighborhood of an αn -ball:

$$\log_2 |\Gamma_{\lceil tn \rceil}(S)| \leq H(\min\{\alpha + t, 1/2\})n + o(n), \quad 0 < t \leq \beta.$$

On any fixed range kept below the equator this is simply the ball exponent $H(\alpha + t)n$. Together with (1), it says that S has the size of an αn -ball and grows like that ball at all scales needed later. We call this *size regularity and pinned growth*.

To see why the conclusion holds, suppose an intermediate neighborhood were exponentially too large. Applying Harper's theorem to that enlarged set for the remaining steps would make the final neighborhood $\Gamma_r(S)$ too large as well. If the comparison radius reaches the equator, the cap (3) supplies the contradiction instead. This snowball argument is Lemma 3.4; the uniform reduction from the hypotheses of Theorem 1.1 to the pinned-growth framework is Proposition 3.5.

2. Every macroscopic coordinate window is generic. Fix $I \subseteq [n]$ with both I and I^c of linear size, and let $X_I = (X_i)_{i \in I}$. This is a discrete random variable in $\{0, 1\}^I$, whose Shannon entropy is

$$H(X_I) = - \sum_{a \in \{0,1\}^I} \mathbb{P}[X_I = a] \log_2 \mathbb{P}[X_I = a].$$

Conclusion of the step. Uniformly for every such window,

$$H(X_I) = H(\alpha)|I| + o(n), \quad H(X_I \mid X_{I^c}) = H(\alpha)|I| + o(n).$$

Thus every macroscopic window carries the expected entropy rate $H(\alpha)$, even after the complementary coordinates have been revealed. This is an entropy statement, not an assertion that the coordinates are independent.

The geometric reason is seen from the fibers. For $z \in \{0, 1\}^{I^c}$ put

$$S_z = \{a \in \{0, 1\}^I : (a, z) \in S\}.$$

Conditionally on $X_{I^c} = z$, the variable X_I is uniform on S_z , and therefore

$$H(X_I | X_{I^c}) = \sum_z \mathbb{P}[X_{I^c} = z] \log_2 |S_z|.$$

Growing different fibers inside the coordinates of I produces disjoint parts of the global neighborhood. A substantial mass of fibers with the wrong exponential size would, by strict concavity of binary entropy, use more than the pinned growth budget. Applying the same argument to the complementary window squeezes both entropy bounds. The precise statement and proof are Theorem 4.1.

3. The minority-branch probability is almost constant. Recall Figure 1. At the random prefix $X_{<t}$,

$$\rho_t = \mathbb{P}[X_t = 1 | X_{<t}], \quad b_t = \min(\rho_t, 1 - \rho_t).$$

The number b_t is the conditional probability that the realized bit takes the less likely branch, or equivalently that X_t disagrees with the canonical-center bit C_t . The parameter α is the relative radius determined by the size of S in (1).

Conclusion of the step. For every fixed tolerance $\eta > 0$,

$$\mathbb{E}_X \#\{t \in [n] : |b_t(X_{<t}) - \alpha| > \eta\} = o(n). \quad (4)$$

In words, a uniformly chosen point encounters only a sublinear expected number of steps at which the less likely branch has probability noticeably different from α . Equivalently, if T is an independent uniformly chosen time, then

$$\mathbb{P}[|b_T(X_{<T}) - \alpha| > \eta] = o(1).$$

By Markov's inequality, a typical point has at most εn such bad steps, for every fixed $\varepsilon > 0$.

Here is the mechanism. Define the deterministic one-step entropy

$$h_t := H(X_t | X_{<t}) = \mathbb{E}_X H(\rho_t) = \mathbb{E}_X H(b_t).$$

Here H denotes Shannon entropy of a random variable, whereas H is the binary entropy function of a scalar probability. The chain rule gives $\sum_t h_t = \log_2 |S| = H(\alpha)n + o(n)$. Applying window genericity to suitably chosen coordinate sets and sorting the h_t shows that all but $o(n)$ of them are close to $H(\alpha)$; this is Theorem 5.1. This average statement alone does not imply (4): different prefixes could have very different values of b_t whose entropies average to the correct number.

To rule this out, quantize the whole path $(b_t)_t$ into narrow bins. A random coordinate window estimates the branch probabilities with $o(n)$ total squared error (Theorem 6.1). Averaging the offset of the quantization grid makes the resulting binned path reconstructible from a random positive-density window with $o(n)$ expected errors. The public-information estimate, Theorem 7.1, then forces this binned path to have entropy $o(n)$; the complete quantization statement is Theorem 8.1.

We may consequently group the points of S into *atoms*, where two points lie in the same atom when they have the same binned path. Almost all points lie in atoms of probability at

least $2^{-o(n)}$. Such an atom loses only $o(n)$ bits of size, remains growth-pinned, and has branch probabilities whose total mean change from the original ones is $o(n)$. For each fixed t , all values of b_t inside one atom lie in the same narrow interval. If this interval were separated from α , monotonicity of H on $[0, 1/2]$ would put all corresponding values $H(b_t)$ strictly on the same side of $H(\alpha)$; their discrepancy could no longer cancel in the average. This contradicts the flat entropy profile of the atom and proves (4). The precise pointwise statement is Theorem 9.1.

4. The canonical center carries only sublinear information. *Conclusion of the step.* For the random center $C = c_S(X)$,

$$H(C) = o(n).$$

Thus the canonical center of a random point can be encoded, with high probability, using only a sublinear number of bits. Lemma 1.3 then turns this entropy bound into a subexponential list of typical centers.

The distinction between magnitude and direction is the key. Step 3 gives $b_t \approx \alpha$, so, outside $o(n)$ exceptional steps, there are two possibilities:

branch probability	minority probability	center bit
$\rho_t \approx \alpha$	$b_t \approx \alpha$	$C_t = 0$
$\rho_t \approx 1 - \alpha$	$b_t \approx \alpha$	$C_t = 1$

The value b_t controls the magnitude of the bias, while C_t records its direction.

Let J be a random window of density $p > 0$. The posterior-variance estimate, Theorem 6.1, says that the possible values of ρ_t after observing the available window prefix have small total variance. Since the two clusters near α and $1 - \alpha$ are separated by a fixed gap, small variance forces almost all posterior mass into one cluster. The window can therefore predict the corresponding bit C_t with small error; the precise two-cluster estimate is Lemma 10.2.

Formally, the center bits are decoded sequentially: when decoding C_t , the previous center bits $C_{<t}$ and the observed prefix $X_{J \cap [1, t]}$ are available. Independence of the random window converts the prediction bound on sampled times into

$$H(C \mid X_J, J) = o(n)$$

for every fixed $p > 0$. This is the sense in which the center is *public information*: after a randomly sampled positive-density window and its locations are revealed, only $o(n)$ additional bits of the center remain to be specified. The window is sampled independently of X , hence of C , and Theorem 4.1 gives $\mathbb{E}_J H(X_J) \leq H(\alpha)pn + o(n)$. Consequently,

$$\begin{aligned} H(C) &= I(C; X_J \mid J) + H(C \mid X_J, J) \\ &\leq \mathbb{E}_J H(X_J) + H(C \mid X_J, J) \leq H(\alpha)pn + o(n). \end{aligned}$$

First keep p fixed and let $n \rightarrow \infty$; then let $p \downarrow 0$ along a diagonal. This gives $H(C) = o(n)$. Theorem 7.1 formalizes the general philosophy that information reconstructible from sparse windows is scarce and is used for the binned path in Step 3. For the canonical center, the direct sequential argument above is Theorem 10.1.

5. A point is close to its own canonical center. *Conclusion of the step.* With probability tending to one,

$$d(X, C) \leq (\alpha + o(1))n = r_{\min}(|S|, n) + o(n).$$

Indeed, put $Z_t = \mathbf{1}[X_t \neq C_t]$. Figure 1 gives the exact conditional mean

$$\mathbb{E}[Z_t \mid X_{<t}] = b_t.$$

Step 3 says that the sum of these conditional means is at most $(\alpha + o(1))n$ for a typical path. Azuma's inequality controls the martingale fluctuations of $\sum_t (Z_t - b_t)$, and therefore controls $d(X, C) = \sum_t Z_t$. The fixed-tolerance estimate is Lemma 11.1; the class-uniform choice of a tolerance tending to zero is Lemma 11.2. Finally, the interior volume calculus, Lemma 2.4, and (1) give $r_{\min}(|S|, n) = \alpha n + o(n)$. This proves the distance conclusion of Proposition 1.2 and completes the outline.

2 Preliminaries: the volume toolbox

All exponents and entropies in Sections 2–3 are base 2; from Section 5 on, the internal computations are base-free (any fixed base), and we write $\ln$ where the natural base matters.

Theorem 2.1 (Harper 1966; black box). *$V(k, r)$ is attained by initial segments of the simplicial order. In particular $V(k, r) = |B(0, t + r)|$ whenever $k = |B(0, t)|$, and $V(\cdot, r)$ is nondecreasing.*

Lemma 2.2 (ball volume, two-sided). *For $0 \leq t \leq n/2$: $|B(0, t)| \leq 2^{H(t/n)n}$; and for t/n in a compact subset of $(0, \frac{1}{2}]$, $|B(0, t)| \geq 2^{H(t/n)n - C_B \log_2 n}$ with a uniform constant C_B .*

Lemma 2.3 (V+: boundary form). *For any $k \geq 1$ and r : $V(k, r) \geq |B(0, t_* + r)|$, where $t_* := \max\{t : |B(0, t)| \leq k\}$.*

Proof. The initial segment of cardinality k contains $B(0, t_*)$; apply monotonicity of Γ_r and Theorem 2.1. $\square$

In plain words. *Lemma 2.3 is the “no assumptions” version of volume calculus: it never mentions entropy and is valid at and beyond the equator. We use it whenever a set might be too small, too big, or too close to radius $n/2$ for asymptotic entropy formulas — a recurring theme is: argue the boundary case with V+ first, then apply interior calculus on the safe range.*

Lemma 2.4 (V: interior volume calculus). *There is $C_V = C_V(\alpha_-, \alpha_+, c_0) \geq 1$ with: for $\alpha \in [\alpha_-, \alpha_+] \subset (0, \frac{1}{2})$, $\alpha + \beta \leq \frac{1}{2} - c_0$, and $|\log_2 k - H(\alpha)n| \leq a$,*

$$|\log_2 V(k, \beta n) - H(\alpha + \beta)n| \leq C_V(a + \log_2 n), \quad |r_{\min}(k, n) - \alpha n| \leq C_V(a + \log_2 n).$$

Proof. Sandwich k between ball cardinalities at radii $\alpha n \mp u$ with $u = O(a + \log_2 n)$, using Lemma 2.2 and the shell ratio $\log_2 \binom{n}{t+1} / \binom{n}{t} = \Theta(1)$ on the compact; then the Harper identity (Theorem 2.1) and monotonicity. $\square$

3 From the hypotheses to growth at every scale

Definition 3.1 (data tuples and classes). A data tuple is $D = (\rho, \delta_c, C_{\text{size}}, [\alpha_-, \alpha_+], \sigma)$ as in Theorem 1.1. $\text{CLASS}(D, n)$ consists of the triples (S, α, β) satisfying (1)–(3), where $\beta = r/n$.

Lemma 3.2 (cap range). Let $\theta := H^{-1}(1 - \delta_c) < \frac{1}{2}$. For every member of $\text{CLASS}(D, n)$: $\alpha + \beta \leq \theta + o(1)$, uniformly in the class.

Proof. By Lemma 2.3 and (3) alone, $|B(0, t_* + \beta n)| \leq V(k, \beta n) \leq |\Gamma_{\beta n}(S)| \leq 2^{(1-\delta_c)n}$, where $t_* \geq \alpha n - C_V(C_{\text{size}}\sigma + \log_2 n)$ by (1) and Lemma 2.2. If $t_* + \beta n \geq n/2$ then $|B(0, t_* + \beta n)| \geq |B(0, \lfloor n/2 \rfloor)| \geq 2^{n - C_B \log_2 n - 1} > 2^{(1-\delta_c)n}$ for large n — contradiction. Hence $t_* + \beta n < n/2$, the interior bound of Lemma 2.2 applies, and $H((t_* + \beta n)/n) \leq 1 - \delta_c + o(1)$; conclude by monotonicity of H on $[0, \frac{1}{2}]$. $\square$

All interior uses of Lemma 2.4 below take $c_0 := (\frac{1}{2} - \theta)/2$.

Definition 3.3 (size regularity and pinned growth). Fix a compact $[q_-, q_+] \subset (0, \frac{1}{2})$, a scale $s_0 > 0$ and a margin $\mu_0 > 0$ with the standing regularity

$$(\text{REG}) \quad q_+ + s_0 \leq \frac{1}{2} - \mu_0,$$

and $\sigma(m) = o(m)$, $\sigma(m) \geq \log_2 m$. Write $\kappa = H(q)$. A set $A \subseteq \{0, 1\}^m$ is size-regular if $|\log_2 |A| - H(q)m| \leq \sigma(m)$ for some $q \in [q_-, q_+]$, and growth-pinned if $|\Gamma_{\lceil \tau m \rceil}(A)| \leq 2^{H(q+\tau)m + \sigma(m)}$ for all $\tau \in (0, s_0]$.

Lemma 3.4 (downward pinning). Let S satisfy (1)–(3) with parameters α, β . Then for every $\tau \in (0, \beta]$,

$$|\Gamma_{\lceil \tau n \rceil}(S)| \leq 2^{H(\min(\alpha+\tau, 1/2))n + o(n)},$$

uniformly over the class.

In plain words. Why does near-optimality at the top radius control every smaller radius? Because growth compounds: if the set overshoot the ball bound at some intermediate scale τ , then running Harper's theorem from that bigger intermediate set for the remaining $\beta - \tau$ steps would overshoot the final bound too — a snowball. The two boundary branches below merely guard the cases where the snowball leaves the sub-equatorial window (then it violates the cap instead).

Proof. Suppose $G_\tau := |\Gamma_{\lceil \tau n \rceil}(S)| \geq 2^{H(\min(\alpha+\tau, 1/2))n + cn}$ for a fixed $c > 0$.

Case (i): $\alpha + \tau \leq \frac{1}{2} - c_0/2$. Let $\alpha' := H^{-1}(\min(H(\alpha + \tau) + c, 1))$, so $\alpha' \geq \alpha + \tau + c'$ with fixed $c' > 0$ (bounded H' on compacts); if the minimum saturates at 1 set $\alpha' := \frac{1}{2}$ and go to the boundary subcase. By Theorem 2.1 applied to $\Gamma_{\lceil \tau n \rceil}(S)$: $|\Gamma_{\beta n}(S)| \geq V(G_\tau, \beta n - \lceil \tau n \rceil)$. If $\alpha' + \beta - \tau \leq \frac{1}{2} - c_0/2$: Lemma 2.4 (on the enlarged compact $[\alpha_-, \frac{1}{2} - c_0/2]$) gives $|\Gamma_{\beta n}(S)| \geq 2^{H(\alpha' + \beta - \tau)n - o(n)} \geq 2^{H(\alpha + \beta)n + c''n - o(n)}$ with fixed $c'' > 0$, contradicting (2) via Lemma 2.4. If $\alpha' + \beta - \tau > \frac{1}{2} - c_0/2$ (boundary subcase): Lemma 2.3 gives $|\Gamma_{\beta n}(S)| \geq |B(0, t_* + \beta n - \lceil \tau n \rceil)|$ with $t_*/n \geq \alpha' - o(1)$, whose exponent is at least $H(\frac{1}{2} - c_0/2) - o(1) > 1 - \delta_c$, contradicting (3).

Case (ii): $\alpha + \tau > \frac{1}{2} - c_0/2$. Then the claimed bound already exceeds the cap: $H(\min(\alpha + \tau, \frac{1}{2})) \geq H(\frac{1}{2} - c_0/2) > 1 - \delta_c$, while $G_\tau \leq |\Gamma_{\beta n}(S)| \leq 2^{(1-\delta_c)n}$ by monotonicity and (3): the supposition fails outright. $\square$

Proposition 3.5 (reduction to pinned growth). *Let $(S, \alpha, \beta) \in \text{CLASS}(D, n)$ and put $s_0 := \rho$, $\theta := H^{-1}(1 - \delta_c)$, $\mu_0 := c_0$. If $\alpha_- > \theta - \rho + \mu_0$ then $\text{CLASS}(D, n)$ is empty for large n (Lemma 3.2) and Theorem 1.1 holds vacuously. Otherwise set $[q_-, q_+] := [\alpha_-, \min(\alpha_+, \theta - \rho + \mu_0)]$; then (REG) holds, since $q_+ + s_0 \leq \theta + \mu_0 \leq \frac{1}{2} - \mu_0$, and $\alpha \in [q_-, q_+]$ for large n by Lemma 3.2. With $q = \alpha$ and $m = n$, the set S is size-regular by (1) and growth-pinned by Lemma 3.4, with the single class-uniform slack $\Sigma(n) := \max(C_{\text{size}}\sigma(n), \sigma_{\text{DP}}(n), \log_2 n) = o(n)$, where σ_{DP} is the slack of Lemma 3.4.*

It therefore suffices to prove the two assertions of Proposition 1.2 for a size-regular, growth-pinned set. Sections 4–10 derive the small entropy of its canonical center, and Section 11 derives the distance estimate. These sections make no further reference to the original class and treat an arbitrary size-regular, growth-pinned set.

4 Pinned growth forces generic coordinate windows

From here on X denotes a uniform random point of A . For $I \subseteq [m]$, the notation X_I means the random restriction of X to the coordinates in I , not merely the set-theoretic projection of A . We write H for Shannon entropy in bits. Thus $H(X_I)$ is the entropy of the probability distribution of X_I . If $J = [m] \setminus I$, then $H(X_I \mid X_J)$ is its conditional entropy after the complementary coordinates have been revealed. Equivalently, for the fibers $A_z = \{a : (a, z) \in A\}$,

$$H(X_I \mid X_J) = \sum_z \mathbb{P}[X_J = z] \log_2 |A_z|,$$

because the conditional distribution on each nonempty fiber is uniform.

Theorem 4.1 (entropy regularity of coordinate windows). *For every fixed $p_{\text{low}} \in (0, \frac{1}{2}]$ there is $\sigma^*(m) = o(m)$, depending only on $(q_-, q_+, s_0, p_{\text{low}}, \sigma)$ — never on the set A or on the window I — such that for every size-regular and growth-pinned A , every $I \subseteq [m]$ with $|I|/m \in [p_{\text{low}}, 1 - p_{\text{low}}]$, and $J = [m] \setminus I$:*

$$|H(X_I) - \kappa|I|| \leq \sigma^*(m), \quad |H(X_I \mid X_J) - \kappa|I|| \leq \sigma^*(m).$$

In plain words. *This is the hinge between geometry and information. Pinning is a statement about neighborhoods; window entropy regularity is a statement about entropies of coordinate projections. The bridge is the fiber picture: fix the coordinates outside I to a value z ; the compatible points form a fiber $A_z \subseteq \{0, 1\}^I$. Two facts do all the work. (1) Disjointness: growing each fiber inside its own copy of $\{0, 1\}^I$ produces disjoint pieces of $\Gamma(A)$ — so all fibers share one pinned budget. (2) Concavity: a sparse fiber (density $< q$) has larger relative growth — balls of smaller radius grow proportionally faster. If a constant fraction of the mass lived in sparse fibers, their joint growth would exceed the budget. Hence the conditional law of X_I given the outside is, for almost all outside values, a fiber of density exactly q : conditional entropy $\kappa|I|$. The unconditional entropy is then squeezed from both sides by counting.*

Proof of Theorem 4.1. Write $p := |I|/m \in [p_{\text{low}}, 1 - p_{\text{low}}]$; radii are rounded up, contributing $O(\log m)$ absorbed in slacks; sums over z range over nonempty fibers. Since X is uniform on A , conditionally on $X_J = z$ the variable X_I is uniform on the fiber $A_z = \{u : (u, z) \in A\}$,

so $H(X_I | X_J) = \mathbb{E}_z \log_2 |A_z|$, with z drawn with probability $|A_z|/|A|$. Define $\alpha_z \in [0, \frac{1}{2}]$ by $|A_z| = 2^{pm H(\alpha_z)}$.

Fiber transfer. For $\theta m \leq s_0 m$:

$$\sum_z |\Gamma_{\theta m}^I(A_z)| \leq |\Gamma_{\theta m}(A)|,$$

because $\Gamma_{\theta m}^I(A_z) \times \{z\} \subseteq \Gamma_{\theta m}(A)$ (flip only I -coordinates) and these slices are disjoint over z .

Sparse fibers are exponentially light. Fix $\zeta \in (0, q_-/2)$ and $\theta := \min(p_{\text{low}}\zeta/2, p_{\text{low}}(\frac{1}{2} - q_+)/2, s_0)$. Let z be sparse: $\alpha_z \leq q - \zeta$. Set $n := pm$, $k := |A_z|$, $r := \lceil \theta m \rceil$, $t_* := \max\{t : |B_n(0, t)| \leq k\}$. Since $|B_n(0, \lfloor \alpha_z n \rfloor)| \leq 2^{nH(\alpha_z)} = k$ (Lemma 2.2, valid down to $\alpha_z = 0$), we get $t_* \geq \lfloor \alpha_z n \rfloor$. By Lemma 2.3 and volume monotonicity,

$$|\Gamma_{\theta m}^I(A_z)| \geq |B_n(0, t_* + r)| \geq |B_n(0, \lfloor \alpha_z n \rfloor + r)| \geq 2^{nH(\alpha_z + \theta/p) - O(\log m)},$$

the last step by Lemma 2.2 at the controlled relative radius $\alpha_z + \theta/p + O(1/m)$, which lies in the fixed compact $[\theta, q - \zeta/2] \subset (0, \frac{1}{2})$ (here $\theta \leq p\zeta/2$ is used). Concavity ($H(x+h) - H(x) \geq h H'(x+h)$) and monotonicity of H' give the relative growth

$$\frac{|\Gamma^I(A_z)|}{|A_z|} \geq 2^{\theta m H'(q - \zeta/2) - O(\log m)}.$$

Summing over sparse fibers, using the fiber transfer, growth-pinned, and $H(q + \theta) - H(q) \leq \theta H'(q)$:

$$\mathbb{P}[\alpha_Z \leq q - \zeta] \leq 2^{-\theta m [H'(q - \zeta/2) - H'(q)] + 2\sigma(m) + O(\log m)} =: 2^{-c(\zeta)\theta m + 2\sigma + O(\log m)},$$

with $c(\zeta) > 0$ on the compact (H' strictly decreasing).

Conditional entropy. The lower bound: $H(X_I | X_J) = \mathbb{E}[pm H(\alpha_Z)] \geq pm H(q - \zeta)(1 - \mathbb{P}[\text{sparse}]) \geq pm\kappa - C\zeta m - o(m)$. The upper bound comes from the complementary split (J in the role of I ; the hypotheses are symmetric): $H(X_J | X_I) \geq (1 - p)m\kappa - C\zeta m - o(m)$, hence

$$H(X_I | X_J) = H(X) - H(X_J) \leq H(X) - H(X_J | X_I) \leq pm\kappa + C\zeta m + \sigma + o(m).$$

Choosing $\zeta(m) \rightarrow 0$ with $\zeta^2 m \gg \sigma(m) + \log m$ (so the sparse mass is $o(1)$ while $\zeta m = o(m)$) yields $|H(X_I | X_J) - \kappa pm| \leq s_{\text{cond}}(m) = o(m)$.

Marginal entropy. $H(X_I) \geq H(X_I | X_J)$, and $H(X_I) = H(X) - H(X_J | X_I) \leq \kappa m + \sigma - [(1 - p)\kappa m - s_{\text{cond}}] = \kappa pm + \sigma + s_{\text{cond}}$. Set $\sigma^* := s_{\text{cond}} + \sigma$. $\square$

Remark (uniformity; subsets). Every constant above depends only on the compact, s_0 , p_{low} and $\sigma(\cdot)$. Moreover if $A' \subseteq A$ satisfies $\log_2 |A'| \geq \kappa m - \sigma_f$, then A' inherits growth-pinned with the same slack ($\Gamma(A') \subseteq \Gamma(A)$) and is size-regular with slack $\max(\sigma, \sigma_f)$; Theorem 4.1 then applies to A' with σ^* computed from $\max(\sigma, \sigma_f)$. Section 9 uses exactly this, with $\sigma_f = \sigma + g(m)$, $g = o(m)$.

5 Flatness of the revelation entropy profile

Reveal the coordinates of X in the fixed order $1, \dots, m$. Let

$$\rho_t := \mathbb{P}[X_t = 1 | X_{<t}], \quad h_t := H(X_t | X_{<t}) = \mathbb{E} H(\rho_t),$$

with entropies in base 2 throughout (as everywhere in this paper; the strong-concavity constant of Section 6 is stated for base 2). By the chain rule $\sum_t h_t = H(X)$.

Theorem 5.1 (flat entropy profile). *Assume window entropy regularity with slack $s(m)$ on the four sets named in the proof. Then for every $\varepsilon > 0$,*

$$\#\{t : |h_t - \kappa| \geq \varepsilon\} \leq \frac{12s(m)}{\varepsilon} + 4.$$

In plain words. *Think of h_t as the rate at which the set spends its entropy budget. The total spend is κm (the set's size), and every window of coordinates must contain spend $\approx \kappa \cdot |\text{window}|$ (window entropy regularity, via the restricted chain rule below). If some steps overspent ($h_t > \kappa + \varepsilon$), the window collecting the biggest spenders would exceed its cap; if some underspent, the window of the smallest spenders would undershoot, forcing — by the conservation of the total — an overshoot elsewhere. A sorting argument makes this exact. The only analytic input is that conditioning on fewer coordinates can only increase conditional entropy.*

Lemma 5.2 (restricted chain rule). *For any I : $H(X_I) = \sum_{t \in I} H(X_t | X_{I \cap [1, t)}) \geq \sum_{t \in I} h_t$.*

Proof. Chain rule inside I in the induced order; each term conditions on a subset of the full past, and $H(X_t | Y) \geq H(X_t | Y, Z)$. $\square$

Proof of Theorem 5.1. Let $k := \lfloor m/4 \rfloor$ and sort: $h_{\pi(1)} \geq \dots \geq h_{\pi(m)}$. Define the deterministic sets $T = \{\pi(1..k)\}$ (top block), $M = \{\pi(k+1..2k)\}$ (second block), $U = \{\pi(1..m-k)\}$, $B_t = [m] \setminus U$ (bottom block). Note every rank in B_t exceeds $2k$. Block-regularity is invoked on T , M , U (sizes in $[m/5, 4m/5]$ for $m \geq 20$; Theorem 4.1 with $p_{\text{low}} = 1/5$) and on $[m]$ (from size-regular directly). With Lemma 5.2 this gives the numeric hypotheses: (i) $\sum_T h_t \leq \kappa k + s$; (ii) $\sum_U h_t \leq \kappa(m-k) + s$; (iii) $\sum_M h_t \leq \kappa k + s$; (iv) $\sum_{[m]} h_t \geq \kappa m - s$.

Floor. By (ii),(iv): $\sum_{B_t} h_t \geq \kappa k - 2s$; every element of T dominates every element of B_t , hence dominates their average: $\min_T h \geq \kappa - 2s/k$.

Up-deviations. Let $u = \#\{h_t \geq \kappa + \varepsilon\}$ and suppose $\varepsilon k > 3s$ (otherwise the bound is vacuous since $12s/\varepsilon + 4 \geq 4k + 4 > m$). If $u > k$, then (i) fails; so all deviants lie in T and (i) with the floor gives $\kappa k + s \geq u(\kappa + \varepsilon) + (k - u)(\kappa - 2s/k)$, i.e. $u \leq 3s/\varepsilon$.

Ceiling and down-deviations. By (iii) the minimal element of M is $\leq \kappa + s/k$; every element of B_t is below it, so $\max_{B_t} h \leq \kappa + s/k$. With $\sum_{B_t} h \geq \kappa k - 2s$ the same computation gives $d := \#\{h_t \leq \kappa - \varepsilon\} \leq 3s/\varepsilon$. $\square$

6 Every window tracks the branching probabilities

For a window W and $t \in W$, the *window observer* of W sees $X_{W \cap [1, t)}$ before step t ; its posterior on the true branching parameter ρ_t (a random variable determined by the full past) has conditional variance $\text{Var}(\rho_t | X_{W \cap [1, t)})$.

Theorem 6.1 (posterior variance budget). *Fix $p_1 \in (0, \frac{1}{2}]$ and assume window entropy regularity with slack $s(m)$ on all W with $|W|/m \in [p_1, 1-p_1]$, plus Theorem 5.1's conclusion with exceptional count $e_\varepsilon(m)$. Then for every such W and every fixed ε :*

$$\sum_{t \in W} \mathbb{E} \text{Var}(\rho_t | X_{W \cap [1, t)}) \leq \frac{\ln 2}{2} (s(m) + \varepsilon |W| + e_\varepsilon(m)).$$

In plain words. *This is where strict concavity earns its keep. When the observer is unsure of the true bias — its posterior on ρ_t has variance v — the bit it is about to see looks like a mixture of coins, and a mixture of coins is more random than the average coin: the observed entropy exceeds the average true entropy by at least $2v/\ln 2$ (Jensen's gap for H , whose second derivative is bounded away from 0). Sum over the window: observed total $= H(X_W) \leq \kappa|W| + s$ (window entropy regularity); true total $\geq (\kappa - \varepsilon)|W| - e_\varepsilon$ (average flatness). The resulting entropy gap is squeezed to $o(m)$. Simultaneously for every window: this is the strength, since each window is just one chain rule.*

Proof. By strong concavity ($|H''| \geq 4/\ln 2$ on $(0, 1)$, minimized at $\frac{1}{2}$): for any random $\rho \in [0, 1]$, $H(\mathbb{E}\rho) - \mathbb{E}H(\rho) \geq \frac{2}{\ln 2} \text{Var}(\rho)$ (Taylor with integral remainder, then expectation). Conditionally on $X_{W \cap [1, t]} = w$, the bit X_t is Bernoulli with parameter $\mathbb{E}[\rho_t | w]$ (tower property), so

$$H(X_t | X_{W \cap [1, t]}) = \mathbb{E}_w H(\mathbb{E}[\rho_t | w]) \geq h_t + \frac{2}{\ln 2} \mathbb{E} \text{Var}(\rho_t | X_{W \cap [1, t]}).$$

Sum over $t \in W$; the left side telescopes to $H(X_W) \leq \kappa|W| + s$; the h_t -sum is $\geq (\kappa - \varepsilon)|W| - e_\varepsilon(m)$ by Theorem 5.1 (deviant steps contribute ≥ 0). Rearrange. $\square$

7 Information visible from every window is scarce

Sample a random window $J \subseteq [m]$ by independent coordinate inclusion with probability p , independently of X .

Theorem 7.1 (public-information bound). *Fix a finite alphabet Σ , $B := |\Sigma| \geq 2$ constant in m , $\kappa \geq 0$, $p_{\text{up}} \in (0, \frac{1}{2})$. Assume $H(X_J) \leq \kappa|J| + s_p(m)$ for every J with $|J| \in [pm/2, 2pm]$, where $s_p(m) = o(m)$ for each fixed $p \in (0, p_{\text{up}})$. Let $\Phi = \Phi(X) \in \Sigma^m$, and suppose that for each fixed p there is a window-computable estimator $\hat{\Phi}_p$ (a function of (X_J, J)) with expected Hamming error $e := \mathbb{E} d_H(\Phi, \hat{\Phi}_p)$. Then*

$$H(\Phi) \leq 2\kappa pm + s_p(m) + F(e, m) + 2m e^{-pm/8}, \quad F(e, m) := m H_2\left(\min\left(\frac{e}{m}, \frac{1}{2}\right)\right) + e \log_2 B.$$

Consequently, if $e = e_p(m) = o(m)$ for each fixed p , then $H(\Phi) = o(m)$.

In plain words. *If every window can reconstruct the string Φ , then Φ is “public”: its information is contained in each window separately. But a window of density p only carries $\approx \kappa pm$ bits in total (window entropy regularity). So the public part of the process is at most κpm bits — for every p — hence $o(m)$. The Fano-type term $F(e, m)$ pays for the reconstruction errors: describing which positions are wrong costs $\approx e \log(m/e)$ bits (an indicator string of density e/m), and correcting them costs $e \log B$. The window-computability requirement is essential: without it one could “estimate” Φ by Φ itself.*

Proof. Coding lemma. If $U, \hat{U} \in \Sigma^m$ are coupled with $\mathbb{E} d_H(U, \hat{U}) \leq e$, then $H(U | \hat{U}) \leq F(e, m)$: let D be the disagreement set and $\mathbf{1}_D$ its indicator string; U is determined by $(\hat{U}, \mathbf{1}_D, U|_D)$; $H(U|_D | \mathbf{1}_D, \hat{U}) \leq \mathbb{E}|D| \log_2 B$; and $H(\mathbf{1}_D) \leq \sum_i H_2(\mathbb{P}[i \in D]) \leq m H_2(\mathbb{E}|D|/m)$ by subadditivity and concavity. Note $m H_2(e/m) = e \log_2(m/e)(1 + o(1)) = o(m)$ whenever $e = o(m)$.

Main chain. With the joint law of (X, J) :

$$H(\Phi) = I(\Phi; (X_J, J)) + H(\Phi | X_J, J) \leq \mathbb{E}_J H(X_J) + H(\Phi | \hat{\Phi}_p, J),$$

using $I(\Phi; J) = 0$ ($J \perp X$) and that $\hat{\Phi}_p$ is a function of (X_J, J) . By Chernoff, $|J| \notin [pm/2, 2pm]$ with probability $\leq 2e^{-pm/8}$, so $\mathbb{E}_J H(X_J) \leq 2\kappa pm + s_p + 2m e^{-pm/8}$. The coding lemma, applied conditionally on J and averaged (the map $e \mapsto F(e, m)$ is concave and nondecreasing), bounds the second term by $F(e, m)$.

Diagonal. Given $c > 0$: take $p := p_{\text{up}}/2$ if $\kappa = 0$, else $p := \min(c/(8\kappa), p_{\text{up}}/2)$; then $2\kappa pm \leq cm/4$ and the remaining terms are $o(m)$ at this fixed, legal p . $\square$

8 The quantized bias magnitudes have sublinear entropy

Let $\text{fold}(x) := \min(x, 1 - x)$ (bias magnitude). Fix a bin width $w > 0$ (a constant; Section 9 takes $w = \varepsilon/4$) and for an offset $o \in [0, w)$ define the *fold field*

$$F^o(X) := \left(\lfloor (\text{fold}(\rho_t) - o)/w \rfloor \right)_{t \leq m} \in \Sigma_w^m,$$

where Σ_w is the (m -independent) set of bins meeting $[0, \frac{1}{2}]$.

Theorem 8.1 (low entropy of the quantized fold field). *There is a deterministic offset $o^* = o^*(m, \text{law})$ and a class-uniform envelope $G_0(m) = o(m)$ with $H(F^{o^*}(X)) \leq G_0(m)$.*

In plain words. By Theorem 6.1, every window tracks ρ_t ; folding and binning are 1-Lipschitz and piecewise constant, so every window can also compute the binned fold trajectory — except near bin boundaries, where a tiny tracking error flips the bin. A uniformly random offset makes each boundary crossing unlikely (probability $|a - b|/w$), so on average over the offset the field is window-computable, and the public-information bound applies — again on average. One then picks the single best offset at the very end. Choosing the offset early (say, per window density p) would break the $p \rightarrow 0$ diagonal — this exact quantifier slip is a pitfall, and the averaged construction below is the repair.

Proof. Fix $p \in (0, \min(\frac{1}{3}, p_{\text{up}}))$ and the window J ; let $\hat{\rho}_t := \mathbb{E}[\rho_t \mid X_{J \cap [1, t]}]$ and let $\hat{F}^o$ bin the folds of $\hat{\rho}$.

Tracking, including out-of-window steps. The quantity $g_t(S) := \mathbb{E}_X(\rho_t - \mathbb{E}[\rho_t \mid X_S])^2$ depends on J only through $S = J \cap [1, t)$, which is *independent* of the event $\{t \in J\}$ under product sampling. Hence, exactly, $p \sum_t \mathbb{E}_J g_t = \mathbb{E}_J \sum_t \mathbf{1}[t \in J] g_t$. Split the right side on $M := \{|J| \in [pm/2, 2pm]\}$: on M , Theorem 6.1 (admissible for $p \leq \frac{1}{3}$) bounds it by $v_p(m)$; on M^c ($\mathbb{P} \leq 2e^{-pm/8}$) crudely by m . Thus

$$\sum_t \mathbb{E}_{J,X}(\rho_t - \hat{\rho}_t)^2 \leq R_p(m) := \frac{v_p(m)}{p} + \frac{2m}{p} e^{-pm/8}, \quad \sum_t \mathbb{E}|\rho_t - \hat{\rho}_t| \leq \sqrt{m R_p(m)} =: L_p^1(m)$$

by Cauchy–Schwarz; $L_p^1(m) = o(m)$ for each fixed p .

Averaged offset error. For a uniform offset and fixed reals a, b : $\mathbb{P}_o[\text{bin}^o(a) \neq \text{bin}^o(b)] \leq |a - b|/w$ (a boundary of the lattice $o + w\mathbb{Z}$ falls between them with at most this probability). Since fold is 1-Lipschitz,

$$\mathbb{E}_o \mathbb{E}_{J,X} d_H(F^o, \hat{F}^o) \leq L_p^1(m)/w =: \bar{e}_p(m) = o(m).$$

Average the public-information bound, then choose the offset. For each fixed o , Theorem 7.1 (quantitative form, arbitrary error) applies to F^o with its window-computable estimator $\hat{F}^o$; averaging over o and using concavity and monotonicity of $F(\cdot, m)$:

$$\mathbb{E}_o H(F^o) \leq 2\kappa pm + s_p(m) + F(\bar{e}_p(m), m) + 2m e^{-pm/8}.$$

The right-hand side is class-uniform; diagonalizing $p(m) \rightarrow 0$ slowly makes it $o(m)$; call it $G_0(m)$. Take $o^* := \arg \min_o H(F^o)$; then $H(F^{o^*}) \leq \mathbb{E}_o H(F^o) \leq G_0(m)$. $\square$

9 Pointwise flatness of the branching magnitudes

Theorem 9.1 (pointwise branching flatness). *For every fixed $\varepsilon \in (0, q_-)$, for every size-regular and growth-pinned A :*

$$\mathbb{E}_X \#\{t : |b_t(X_{<t}) - q| \geq \varepsilon\} = o(m), \quad b_t := \text{fold}(\rho_t) = \min(\rho_t, 1 - \rho_t),$$

with slack uniform over the class.

In plain words. *The flat entropy profile says the mean step entropy is κ ; it does not exclude a bimodal world where some trajectories see fair coins ($\rho = \frac{1}{2}$) and others see frozen bits ($\rho \in \{0, 1\}$), averaging to κ . Such bimodality would wreck everything downstream (the “center” would sit at distance $\approx \kappa m/2 \neq qm$). The repair has three moves. (1) Atoms: group points by their binned fold trajectory. Since the fold field has $o(m)$ entropy, heavy atoms — of mass $2^{-o(m)}$ — carry almost all mass. (2) Heredity + tilt: a heavy atom is still size-regular and growth-pinned, hence window-regular, so its own profile is flat on average; and conditioning on the atom bends branchings only slightly: the total KL cost equals $\log_2(1/\text{mass}) = o(m)$, which Pinsker converts to an ℓ_1 bound. (3) Same-bin separation: inside an atom, the fold at step t is trapped in one bin of width $\varepsilon/4$. If even one trajectory of the atom strays ε from q at step t , the entire bin sits at distance $\geq 3\varepsilon/4$ from q — and, crucially, on one side of q , so the step entropies cannot cancel around κ : the atom’s average step entropy at t is separated from κ by a fixed gap, contradicting (2). An earlier version of this step tried to invert H through its global modulus of continuity; that is quantitatively false (the modulus near 0 beats the local gap at q); the one-sided same-bin argument is the repair.*

Proof. Let $F = F^{o^*}$ be the fold field with $w = \varepsilon/4$, and $G_0(m)$ its envelope (Theorem 8.1). Set $g := \max(\sqrt{G_0(m)m}, m^{2/3}) = o(m)$, a deterministic, class-uniform threshold. Call ϕ heavy if $\mu(\phi) := |A_\phi|/|A| \geq 2^{-g}$ where $A_\phi = \{x \in A : F(x) = \phi\}$. By Markov on $-\log_2 \mu(F(X))$ (mean $\leq G_0$): $\mathbb{P}[F(X) \text{ heavy}] \geq 1 - G_0/g = 1 - o(1)$.

Bin property. For $x \in A_\phi$ and every t : $\text{fold}(\rho_t(x_{<t})) \in I_{t,\phi} := \text{bin}(\phi_t) \cap [0, \frac{1}{2}]$, an interval of length $\leq w$ (nonempty as it contains a realized fold).

Heavy atoms are window-regular. $|A_\phi| \geq 2^{\kappa m - \sigma - g}$ and $\Gamma(A_\phi) \subseteq \Gamma(A)$: by Remark 4, Theorem 4.1 applies to A_ϕ with slack computed from $\sigma + g = o(m)$ — the same for all heavy atoms.

KL tilt. Let $\rho_t^\phi := \mathbb{P}[X_t = 1 \mid X_{<t}, F(X) = \phi]$ be the branchings of the atom’s own tree (X^ϕ uniform on A_ϕ). Conditioning on an event of probability $\mu(\phi)$ costs $\text{KL}(\mathbb{P}[\cdot \mid \phi] \parallel \mathbb{P}) = \log_2(1/\mu(\phi)) \leq g$; the chain rule for relative entropy splits this along the filtration into per-step

Bernoulli divergences at realized prefixes; Pinsker ($\text{TV} \leq \sqrt{\text{KL}/2}$; TV of Bernoullis is $|\rho_t^\phi - \rho_t|$), Jensen and Cauchy–Schwarz give

$$\sum_t \mathbb{E}_{X^\phi} |\rho_t^\phi - \rho_t| \leq \sqrt{mg/2} = o(m).$$

By uniform continuity of H (concave modulus ω), put

$$B_m := \sum_t \mathbb{E}_{X^\phi} |H(\rho_t^\phi) - H(\rho_t)| \leq m \omega(\sqrt{mg/2}/m) = o(m).$$

Markov with threshold $\delta_m := \sqrt{B_m/m} \rightarrow 0$ gives a set $T_{\text{bad}}(\phi)$ of at most $\sqrt{B_m m} = o(m)$ coordinates outside which

$$|\mathbb{E}_{X^\phi} H(\rho_t) - \mathbb{E}_{X^\phi} H(\rho_t^\phi)| \leq \delta_m. \quad (\text{BR})$$

Same-bin separation. Define the compact gap

$$\Delta_\varepsilon := \inf \{ |H(y) - H(q')| : q' \in [q_-, q_+], y \in [0, \frac{1}{2}], |y - q'| \geq \frac{3\varepsilon}{4} \} > 0.$$

Apply Theorem 5.1 to X^ϕ with $\varepsilon' := \Delta_\varepsilon/4$: all but $o(m)$ coordinates satisfy $|\mathbb{E}_{X^\phi} H(\rho_t^\phi) - \kappa| \leq \varepsilon'$; call these coordinates *profile-good* for the atom.

Claim. If $t \notin T_{\text{bad}}(\phi)$, t is profile-good, and $\delta_m \leq \Delta_\varepsilon/4$, then *every* $x \in A_\phi$ has $|b_t(x) - q| < \varepsilon$. Indeed, suppose some x_0 strays: $|b_t(x_0) - q| \geq \varepsilon$. All folds of the atom at step t lie in the single interval $I_{t,\phi}$ of length $\leq \varepsilon/4$; hence every $y \in I_{t,\phi}$ has $|y - q| \geq 3\varepsilon/4$, and — the interval being too short to straddle q — all on *one side*. By strict monotonicity of H on $[0, \frac{1}{2}]$, all values $H(y)$ lie on one side of κ at distance $\geq \Delta_\varepsilon$; hence the *average* obeys $|\mathbb{E}_{X^\phi} H(b_t) - \kappa| \geq \Delta_\varepsilon$. But (BR) and profile-goodness give $|\mathbb{E}_{X^\phi} H(b_t) - \kappa| \leq \delta_m + \varepsilon' \leq \Delta_\varepsilon/2$ (note $H(\rho_t) = H(b_t)$ by fold symmetry): contradiction.

Assembly.

$$\begin{aligned} \mathbb{E}_X \#\{t : |b_t - q| \geq \varepsilon\} &= \sum_{\phi \text{ heavy}} \mu(\phi) \mathbb{E}_{X^\phi} \#\{\cdots\} \\ &\quad + \sum_{\phi \text{ light}} \mu(\phi) \mathbb{E}_{X^\phi} \#\{\cdots\} \\ &\leq o(m) + \mathbb{P}[\text{light}] \cdot m = o(m), \end{aligned}$$

the per-atom count being $o(m)$ uniformly over heavy atoms (all slacks are functions of (G_0, σ, m) only). $\square$

10 The canonical center has sublinear entropy

For $x \in A$, define the *canonical-center map* $c_A : A \rightarrow \{0, 1\}^m$ by

$$c_A(x)_t := \mathbf{1}[\rho_t(x_{<t}) > 1/2].$$

For the uniform random point X , put $C = c_A(X)$ and write C_t for its t th coordinate. Thus C is a $\{0, 1\}^m$ -valued random variable. Every entropy involving C below is the Shannon entropy of its induced probability distribution, in the same sense as in Subsection 1.3.

Theorem 10.1 (small entropy of the canonical center). *Under window entropy regularity, the posterior variance budget, and pointwise branching flatness, $H(C) = o(m)$, with class-uniform slack.*

In plain words. After Theorem 9.1, the process is a q -biased coin at almost every step; the only freedom left is the direction of the bias — the center bit C_t . Directions could a priori carry full entropy (a ball around an unknown center has exactly this structure). Two forces cap them. First, pointwise branching flatness puts the branching parameter in two clusters, near q and near $1 - q$, separated by a fixed gap; the posterior variance theorem shows that each window's total posterior variance is tiny; and a two-cluster distribution with tiny variance must concentrate on one cluster — so the window knows the direction at almost every step (two-cluster lemma). Second, the public-information mechanism of Section 7: something every window can compute has at most κpm bits for every p . The formal vehicle is the novelty $\nu_t(J) = H(C_t \mid C_{<t}, X_{J \cap [1,t)})$ and an exact independence identity for random windows.

Lemma 10.2 (two-cluster). Fix $\varepsilon \in (0, (1 - 2q_+)/2)$, a time t , and a realization w of $X_{J \cap [1,t)}$. Let $\beta_w := \mathbb{P}[|b_t - q| > \varepsilon \mid w]$, $\gamma_w := \mathbb{P}[\rho_t > \frac{1}{2} \mid w]$, $\delta_w := \min(\gamma_w, 1 - \gamma_w)$, and $\text{gap} := 1 - 2q_+ - 2\varepsilon > 0$. Then

$$\delta_w \leq \frac{4}{\text{gap}^2} \text{Var}(\rho_t \mid w) + 2\beta_w.$$

Proof. Outside the branching-bad event, $\rho_t \in [q - \varepsilon, q + \varepsilon] \cup [1 - q - \varepsilon, 1 - q + \varepsilon]$ — two intervals at distance $\geq \text{gap}$, with conditional masses a, b , $a + b \geq 1 - \beta_w$. From $\text{Var} = \frac{1}{2} \mathbb{E}(\rho - \rho')^2$ restricted to cross-cluster pairs: $\text{Var}(\rho_t \mid w) \geq ab \text{gap}^2$. If $\beta_w \geq \frac{1}{2}$ the claim is trivial; otherwise $\max(a, b) \geq \frac{1}{4}$, so $\min(a, b) \leq 4\text{Var}/\text{gap}^2$; and $|\gamma_w - b| \leq \beta_w$ yields $\delta_w \leq \min(a, b) + 2\beta_w$. $\square$

Proof of Theorem 10.1. Fix $\varepsilon \in (0, \min(q_-, (1 - 2q_+)/2))$ and $p \in (0, \frac{1}{3}]$; sample J by independent p -inclusion. On the event $M = \{|J| \in [pm/2, 2pm]\}$, Theorem 6.1 gives $\sum_{t \in J} \mathbb{E} \text{Var}(\rho_t \mid X_{J \cap [1,t)}) \leq v(m)$ for every macroscopic realization of J . Put $\beta_t := \mathbb{P}[|b_t - q| > \varepsilon]$. Theorem 9.1 gives a class-uniform bound $\sum_t \beta_t \leq b(m) = o(m)$; here β_t does not depend on J . Averaging Lemma 10.2 over w and summing over $t \in J$ therefore yields

$$\sum_{t \in J} \bar{\delta}_t(J) \leq \frac{4}{\text{gap}^2} v(m) + 2b(m) =: D(m) = o(m).$$

Let $\nu_t(J) := H(C_t \mid C_{<t}, X_{J \cap [1,t)})$, a functional of $S_t = J \cap [1, t)$ alone. By folding and Jensen, $\nu_t(J) \leq H(\bar{\delta}_t(J))$, and by concavity over $t \in J$,

$$\sum_{t \in J} \nu_t(J) \leq \beta_p^*(m) := \sup_{j \in [pm/2, 2pm]} j H\left(\min\left(\frac{D(m)}{j}, \frac{1}{2}\right)\right) = o(m).$$

Independence trick. Under product sampling, $S_t \perp \{t \in J\}$, so $\mathbb{E}_J[\nu_t \mathbf{1}[t \in J]] = p \mathbb{E}_J[\nu_t]$. On M^c use the crude bound $\sum_{t \in J} \nu_t \leq m$ and the Chernoff estimate $\mathbb{P}(M^c) \leq 2e^{-pm/8}$. Averaging the per-realization bound therefore gives $\sum_t \mathbb{E}_J \nu_t \leq (\beta_p^*(m) + o(m))/p$. *Compression.* $H(C \mid X_J) = \sum_t H(C_t \mid C_{<t}, X_J) \leq \sum_t \nu_t(J)$ (conditioning on the whole window only helps). *Assembly.* For every realization, $H(C) = I(C; X_J) + H(C \mid X_J) \leq H(X_J) + H(C \mid X_J)$; take $\mathbb{E}_J$, apply window entropy regularity and the two displays:

$$H(C) \leq \kappa pm + s_p(m) + \frac{\beta_p^*(m) + o(m)}{p} + o(1).$$

Fixed p then $p \rightarrow 0$ (diagonal) gives $H(C) = o(m)$. $\square$

11 Completing the canonical-center estimate

We now use only two conclusions of the preceding sections. Uniformly over all size-regular, growth-pinned sets in the fixed class, retain the notation $C = c_A(X)$ and write

$$H(C) \leq g(m), \quad g(m) = o(m), \quad (5)$$

$$\mathbb{E} \#\{t : |b_t - q| \geq \eta\} \leq b_\eta(m), \quad b_\eta(m) = o(m) \quad \text{for every fixed } \eta > 0. \quad (6)$$

The function b_η may depend on the fixed threshold η . We keep this dependence explicit; the diagonal lemma below is precisely what permits the threshold to tend to zero.

Lemma 11.1 (distance from the canonical center). *For every fixed $\eta \in (0, 1/2 - q_+)$,*

$$\mathbb{P}[d(X, C) > (q + 3\eta)m] \leq \frac{b_\eta(m)}{\eta m} + e^{-\eta^2 m/2}.$$

Proof. Put $Z_t = \mathbf{1}[X_t \neq C_t]$. Given the full past $X_{<t}$,

$$\mathbb{E}[Z_t \mid X_{<t}] = \min(\rho_t, 1 - \rho_t) = b_t.$$

By (6) and Markov's inequality,

$$\mathbb{P}[\#\{t : b_t > q + \eta\} > \eta m] \leq \frac{b_\eta(m)}{\eta m}.$$

The process

$$M_T = \sum_{t \leq T} (Z_t - b_t)$$

is a martingale with increments in $[-1, 1]$, so Azuma's inequality gives $\mathbb{P}[M_m \geq \eta m] \leq e^{-\eta^2 m/2}$. Outside the two exceptional events,

$$d(X, C) = \sum_t Z_t \leq \sum_t b_t + \eta m \leq (q + 3\eta)m.$$

The last inequality uses $b_t \leq q + \eta$ at all but at most ηm steps and the crude bound $b_t \leq 1/2$ at the remaining steps. $\square$

Lemma 11.2 (uniform vanishing threshold). *There is a deterministic sequence $\eta_m \downarrow 0$, depending only on the fixed class data, such that*

$$\sup_A \mathbb{P}[d(X, C) > (q + 3\eta_m)m] \longrightarrow 0,$$

where the supremum is over all size-regular, growth-pinned $A \subseteq Q_m$ and all their admissible parameters q .

Proof. Choose any decreasing sequence $\eta_j \downarrow 0$ inside $(0, 1/2 - q_+)$. For each fixed j , the class-uniform estimate (6) and Lemma 11.1 give a threshold N_j such that, for every $m \geq N_j$ and every admissible set,

$$\mathbb{P}[d(X, C) > (q + 3\eta_j)m] \leq 2^{-j}.$$

Increase the N_j if necessary so that $N_1 < N_2 < \dots$. Set $\eta_m = \eta_j$ when $N_j \leq m < N_{j+1}$, and define it arbitrarily for $m < N_1$. Then $j = j(m) \rightarrow \infty$, hence $\eta_m \rightarrow 0$ and the displayed probability is at most $2^{-j(m)} \rightarrow 0$, uniformly over the class. $\square$

Proof of Proposition 1.2. Let S satisfy (1)–(3). By Proposition 3.5, with $m = n$ and $q = \alpha$, it is size-regular and growth-pinned with one class-uniform sublinear slack. Consequently Theorems 4.1, 5.1, 6.1, 7.1, 8.1, 9.1, and 10.1 apply with envelopes depending only on the fixed data. Theorem 10.1 gives a deterministic $g(n) = o(n)$ such that $H(C) \leq g(n)$.

Lemma 11.2 supplies $\eta_n \downarrow 0$ and a deterministic $e(n) \rightarrow 0$ such that

$$\mathbb{P}[d(X, C) > (\alpha + 3\eta_n)n] \leq e(n)$$

uniformly over the original class. By (1) and the interior volume calculus, Lemma 2.4, there is a deterministic $v(n) = o(n)$, again uniform over the class, such that

$$|r_{\min}(|S|, n) - \alpha n| \leq v(n).$$

Set $u(n) = 3\eta_n n + v(n) + 1$. Then $u(n) = o(n)$ and

$$(\alpha + 3\eta_n)n \leq r_{\min}(|S|, n) + u(n).$$

The preceding probability estimate therefore proves the distance conclusion of Proposition 1.2. Enlarging g, u, e harmlessly in finitely many small dimensions gives the stated uniform total form. $\square$

12 Verification status

The main theorem, Harper’s theorem, and the geometric, entropy, and martingale ingredients used in the proof have been formalized in Lean 4/Mathlib. The paper proof is presented in a slightly simplified form for readability.

References

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